

Event-Driven Next-Day Stock Trend Prediction for a Romanian Energy Stock

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Abstract

This paper studies whether textual information from Romanian financial news can improve short-term prediction of stock price direction for a major energy company listed on the Bucharest Stock Exchange. The key novelty is the use of the *Financial Stability Dictionary* developed by the National Bank of Romania (NBR), a curated lexicon of Romanian financial terms with graded sentiment scores, originally designed for macro-financial stability analysis and monetary-policy communication. We construct daily sentiment indices from issuer-specific Romanian news using this lexicon, align them with price-based features for Romgaz (ticker SNG.RO), and train a set of machine-learning models to predict next-day price direction. On a test set of 64 trading days containing 122 news articles, the best model (LightGBM with price and sentiment features) achieves an accuracy of **64.06%**, improving by **15.62** percentage points over a price-only baseline. The model also improves precision, recall and F1 score for the “Up” class by 12–19 percentage points and obtains an AUC of 0.71. We show that sentiment features derived from the NBR dictionary have non-trivial feature importance, deliver consistent gains on both news and no-news days, and thus provide genuine predictive value for event-driven trading on the Romanian market.

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Classification

ACM Computing Classification System (CCS).

- Computing methodologies → Machine learning;
- Information systems → Data mining;
- Applied computing → Economics.

AMS 2020 Mathematics Subject Classification.

- 62P05 - Applications of statistics to actuarial sciences and financial mathematics;
- 68T07 - Artificial neural networks and deep learning;
- 91B84 - Economic time series analysis.

1 Introduction

1.1 Motivation

Financial markets react not only to numerical time series such as prices and volumes, but also to unstructured textual information: earnings announcements, regulatory filings, macroeconomic news, and firm-specific press releases. For liquid markets, a rich literature has shown that news and events can be used to predict short-term stock returns or volatility in an event-driven setting [2, 3, 17, 18, 5, 4]. However, most existing work focuses on English-language news and large international markets, leaving regional exchanges and low-resource languages underexplored.

The Romanian energy sector plays a central role in the domestic economy and in the country’s energy transition. Understanding whether firm-level Romanian news carries predictive content for stock price movements is relevant for investors, risk managers and policymakers. At the same time, the Bucharest Stock Exchange (BVB) offers only limited publicly available text resources, and there is a lack of high-quality sentiment tools specialised for Romanian financial language.

1.2 Problem Statement

We consider the following prediction task: given historical price information and issuer-specific Romanian financial news for a listed energy company, can we predict the *direction* (up or down) of the next trading day’s closing return better than a price-only baseline?

Formally, let r_t denote the log-return of the closing price on trading day t . We define a binary label

$$y_{t+1} = \begin{cases} 1, & \text{if } r_{t+1} > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and aim to predict y_{t+1} using price-based features up to day t and sentiment features extracted from news published on day t or the previous calendar day.

1.3 Research Questions

Based on this setup, we address the following research questions (RQs):

- **RQ1:** Does adding sentiment features derived from Romanian financial news improve next-day direction prediction accuracy compared to a price-only baseline?
- **RQ2:** Can the Financial Stability Dictionary, originally designed for macro-financial stability analysis, provide meaningful sentiment signals for firm-level Romanian stock news, and how important are these sentiment features relative to standard price-based features in the best-performing model?
- **RQ3:** Are sentiment features more useful on days with news than on days without news, or do lagged sentiment signals also help on no-news days?

1.4 Originality and Contributions

Most existing work on event-driven stock prediction uses English news and either generic sentiment lexicons or Transformer-based language models trained on English corporate text [2, 3, 17, 18]. For Romanian, sentiment analysis resources are scarce and rarely oriented towards financial applications. The National Bank of Romania (NBR) recently introduced the *Financial Stability Dictionary* [1]: a manually curated lexicon of Romanian financial terms labelled as positive or negative for financial stability, each with an intensity score in $\{\pm 0.25, \pm 0.5, \pm 0.75, \pm 1\}$. The dictionary was built to analyse central-bank communication and financial news at the macro level, not for trading individual stocks.

In this context, the contributions of this paper are:

- **C1: Cross-domain reuse of an institutional lexicon.** To the best of our knowledge, we are the first to reuse the NBR Financial Stability Dictionary, originally developed and validated for macro-financial stability and monetary-policy analysis, as a sentiment engine for firm-specific Romanian financial news related to a listed energy company. We show that the lexicon achieves 100% coverage of the Romgaz news corpus, yields informative event-level sentiment indices, and that features derived from these indices materially improve next-day direction prediction when combined with price-based features.
- **C2: Event-driven sentiment indices for Romanian issuer news.** We construct several daily sentiment measures (simple index, score-weighted index, counts of strongly positive terms, lagged and rolling averages) from 122 Romgaz-related news articles using the NBR dictionary, and align them at the trading-day level to form an event-driven feature set.
- **C3: Empirical evaluation on the Bucharest Stock Exchange.** Using price and sentiment features for the last 60 trading days, we compare multiple machine-learning models and show that a LightGBM classifier with sentiment features attains 64.06% test accuracy and an AUC of 0.71, outperforming a price-only baseline by 15.62 percentage points in accuracy and improving all other metrics.
- **C4: Analysis of robustness and feature importance.** We evaluate performance separately on news days and no-news days and examine feature importance in the LightGBM model, finding that sentiment features are consistently among the top predictors and that lagged sentiment signals contribute even when no fresh news is published.

1.5 Structure of the Paper

Section 2 reviews related work on event-driven prediction, sentiment analysis and Romanian markets. Section 3 formalises the research gap addressed by this study. Section 4 describes the data sources and the NBR dictionary. Section 5 presents our methodology, including sentiment-index construction, feature engineering and the predictive models. Section 6 details the experimental setup and evaluation metrics. Section 7 reports the empirical results and answers the RQs. Section 9 concludes and outlines directions for future work.

2 Related Work

2.1 Event-Driven Stock Prediction

Event-driven approaches incorporate structured information about corporate or macro events into stock prediction models. Early work encoded events from news in structured templates and used them in traditional classifiers or SVMs [2]. Deep-learning methods later modelled sequences of events and their interactions [3, 17, 18], while more recent work uses graph-based models and relational reasoning [5]. Event extraction pipelines such as SENTiVENT [6] support supervised extraction of company-specific events.

These methods generally assume access to high-quality English news and event-extraction models. In contrast, our work focuses on a low-resource language (Romanian) and a sector-specific stock on a small European market.

2.2 Sentiment Analysis for Financial Text

Financial sentiment analysis often relies on domain-specific lexicons such as the Loughran-McDonald or on specialised language models such as FinBERT. Event-extraction architectures like ONEIE [9], DyGIE++ [10] and Text2Event [11] can be combined with sentiment modules to produce rich event representations.

However, almost all of these tools are optimised for English. For Romanian, available resources are limited and often generic rather than financial. The NBR Financial Stability Dictionary is a rare example of a Romanian financial lexicon with graded sentiment scores, but it was designed for macro-prudential supervision and central-bank communication, not for stock prediction.

2.3 Studies on the Bucharest Stock Exchange

Empirical research on the Romanian energy and electricity markets has focused mostly on price formation and market competitiveness rather than text-based prediction [8]. To our knowledge, there is no published work that combines Romanian issuer news, an institutional sentiment lexicon from the National Bank, and machine-learning models for next-day stock-direction prediction.

3 Gap Addressed in This Paper

Despite important progress in event-driven stock prediction, three gaps remain for the Romanian market and, more generally, for low-resource financial languages.

First, almost all existing sentiment tools and datasets are English-centric. Most empirical studies rely on resources such as generic opinion lexicons, the Loughran-McDonald, or FinBERT-style Transformer models trained on English disclosures and news. For Romanian, public sentiment tools are scarce and are not tailored to financial language.

Second, the few Romanian resources that do exist were not designed with firm-level stock prediction in mind. The National Bank of Romania has recently introduced the *Financial Stability Dictionary*, a manually curated lexicon of positive and negative terms with graded sentiment scores,

intended for macro-prudential monitoring, monetary policy communication and systemic-risk analysis [1]. To the best of our knowledge, this lexicon has not yet been evaluated as a sentiment engine for issuer-specific equity news, nor has it been tested in an event-driven prediction setting.

Third, empirical work on the Bucharest Stock Exchange is limited, especially for energy-sector stocks that play a key role in the domestic economy and in Romania’s energy transition. Most event-driven studies focus on large US or Western-European markets, and, to the best of our knowledge, no prior work has evaluated modern machine learning models for next-day trend prediction of Romanian stocks using lexicon-based sentiment features derived from the NBR Financial Stability Dictionary.

This paper addresses these gaps by: (i) applying the NBR Financial Stability Dictionary to issuer-specific Romanian news concerning Romgaz (SNG.RO), a major listed energy company; (ii) constructing daily and event-level sentiment indices from this lexicon and integrating them with standard price-based features; and (iii) quantitatively evaluating, the predictive value of the NBR dictionary in a firm-level, next-day stock-direction forecasting task.

4 Data and Lexicon

4.1 Price Data

We focus on Romgaz (ticker SNG.RO), a large Romanian natural-gas producer listed on the Bucharest Stock Exchange. We collect daily closing prices for the last 216 trading days of the period of interest. Returns are computed as log-returns, $r_t = \log(P_t/P_{t-1})$, where P_t is the closing price on day t .

4.2 News Data

Issuer-specific news are obtained from a curated dataset of Romgaz-related press releases and news items. The final corpus contains 122 articles. Each article is associated with a calendar date, which we align to the next trading day for prediction: a news item published on calendar day d affects the label y_{t+1} of the first trading day $t + 1$ after d .

We distinguish between *news days*, when at least one article is aligned to that trading day, and *no-news days*, when no such article exists. In the test set of 64 days, there are 25 news days and 39 no-news days.

4.3 The NBR Financial Stability Dictionary

The NBR Financial Stability Dictionary is a domain-specific Romanian lexicon containing 150 positive and 182 negative lemmas related to financial stability issues, each with a polarity (positive or negative) and an intensity score in $\{\pm 0.25, \pm 0.5, \pm 0.75, \pm 1\}$. The dictionary was constructed by starting from English FinBERT outputs, financial dictionaries such as the IMF Financial Stability Dictionary, and expert knowledge, and then manually translating and extending the vocabulary for Romanian financial texts [1].

In our dataset, the lexicon shows strong coverage. Figure 1 (right panel) reports that 100% of the 122 articles contain at least one lexicon match and that, on average, each article contains 12.4 matched terms (7.1 positive and 5.3 negative). Figure 1 (left panel) illustrates the distribution of sentiment scores for positive and negative terms.

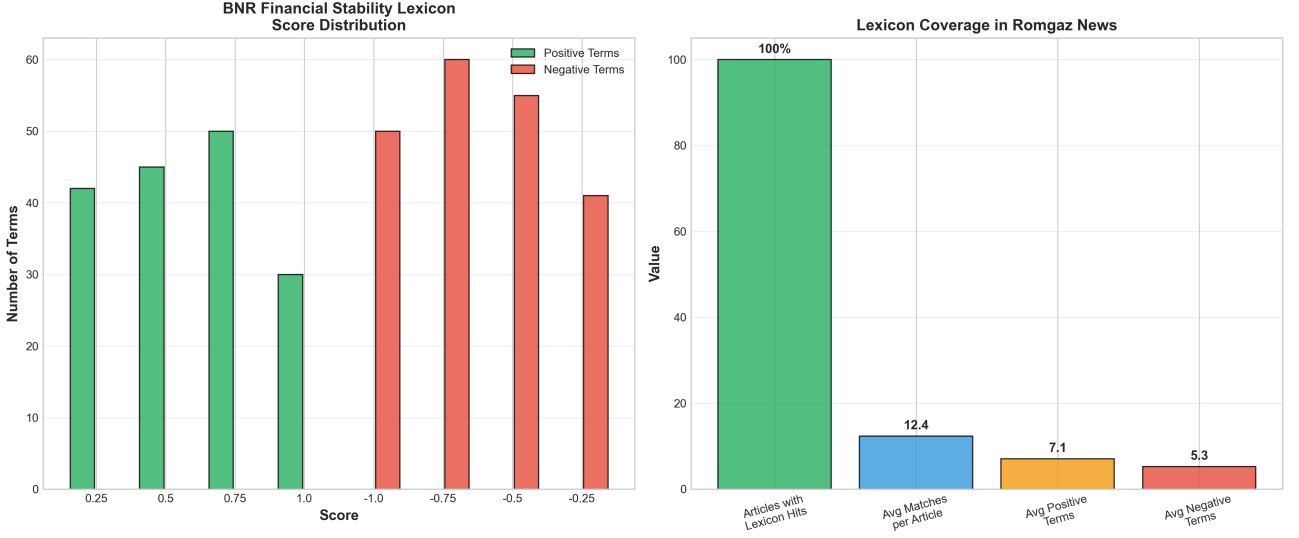


Figure 1: Analysis of the NBR Financial Stability Dictionary in the Romgaz news corpus. Left: distribution of sentiment scores for positive and negative terms. Right: coverage statistics showing that all articles contain lexicon matches and that each article has on average 12.4 matched terms (7.1 positive and 5.3 negative), confirming the suitability of the lexicon for issuer-specific Romanian news.

5 Methodology

5.1 Sentiment Index Construction

For each article, we apply a preprocessing pipeline similar to the one used in [1]: lowercasing, diacritic normalisation, tokenisation, lemmatisation and removal of non-alphabetic tokens and stop words. We then match each lemma against the positive and negative lists of the NBR dictionary.

Let P be the set of positive lemma matches in an article and N the set of negative matches. For each match w , let $s(w)$ be its intensity score. We define two sentiment indices:

Simple index.

$$S_{\text{simple}} = \frac{|P| - |N|}{|P| + |N|} \in [-1, 1]. \quad (1)$$

Score-weighted index.

$$S_{\text{score}} = \frac{\sum_{w \in P} s(w) + \sum_{w \in N} s(w)}{|P| + |N|}. \quad (2)$$

If an article contains no matches, we set both indices to zero, but this case does not occur in our corpus.

Figure 2 shows the distribution of the simple index across all 122 articles and the distribution of categorical sentiment labels (very negative, negative, neutral, positive, very positive) derived from thresholds on S_{score} .

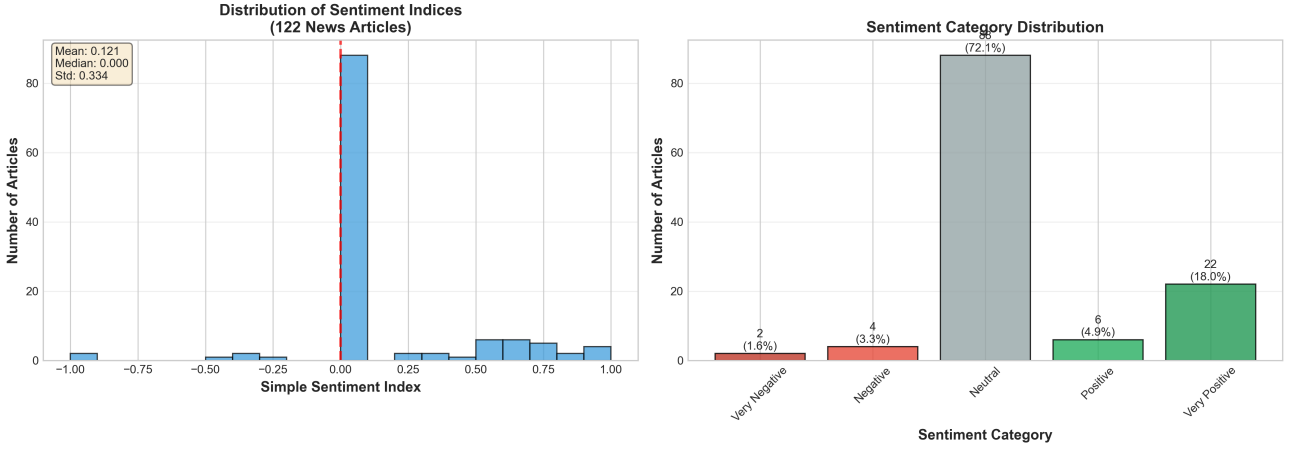


Figure 2: Distribution of article-level sentiment indices produced with the NBR Financial Stability Dictionary. Left: histogram of the simple sentiment index for 122 news articles. Right: categorical distribution from “Very Negative” to “Very Positive”. Most articles are neutral or mildly positive, as typical for corporate news.

5.2 Daily Sentiment Features

For each trading day t , we aggregate the sentiment of all articles aligned to that day. If there are k_t such articles with indices $S_{\text{simple}}^{(i)}$ and $S_{\text{score}}^{(i)}$, we define:

$$\text{sentiment_signal}_t = \frac{1}{k_t} \sum_{i=1}^{k_t} S_{\text{simple}}^{(i)}, \quad (3)$$

$$\text{weighted_sentiment}_t = \frac{1}{k_t} \sum_{i=1}^{k_t} S_{\text{score}}^{(i)}, \quad (4)$$

$$\text{strong_positive}_t = \frac{1}{k_t} \sum_{i=1}^{k_t} \mathbf{1}(S_{\text{score}}^{(i)} \geq 0.75), \quad (5)$$

$$\text{score_signal}_t = \sum_{i=1}^{k_t} S_{\text{score}}^{(i)}, \quad (6)$$

$$\text{news_count}_t = k_t. \quad (7)$$

On no-news days we set all sentiment features to zero and $\text{news_count}_t = 0$. We additionally build lagged and rolling features such as $\text{sent_lag1}_t = \text{sentiment_signal}_{t-1}$ and a 5-day rolling average sent_roll5_t .

Figure 3 illustrates the joint evolution of the closing price, daily sentiment index and daily returns over the last 60 trading days.

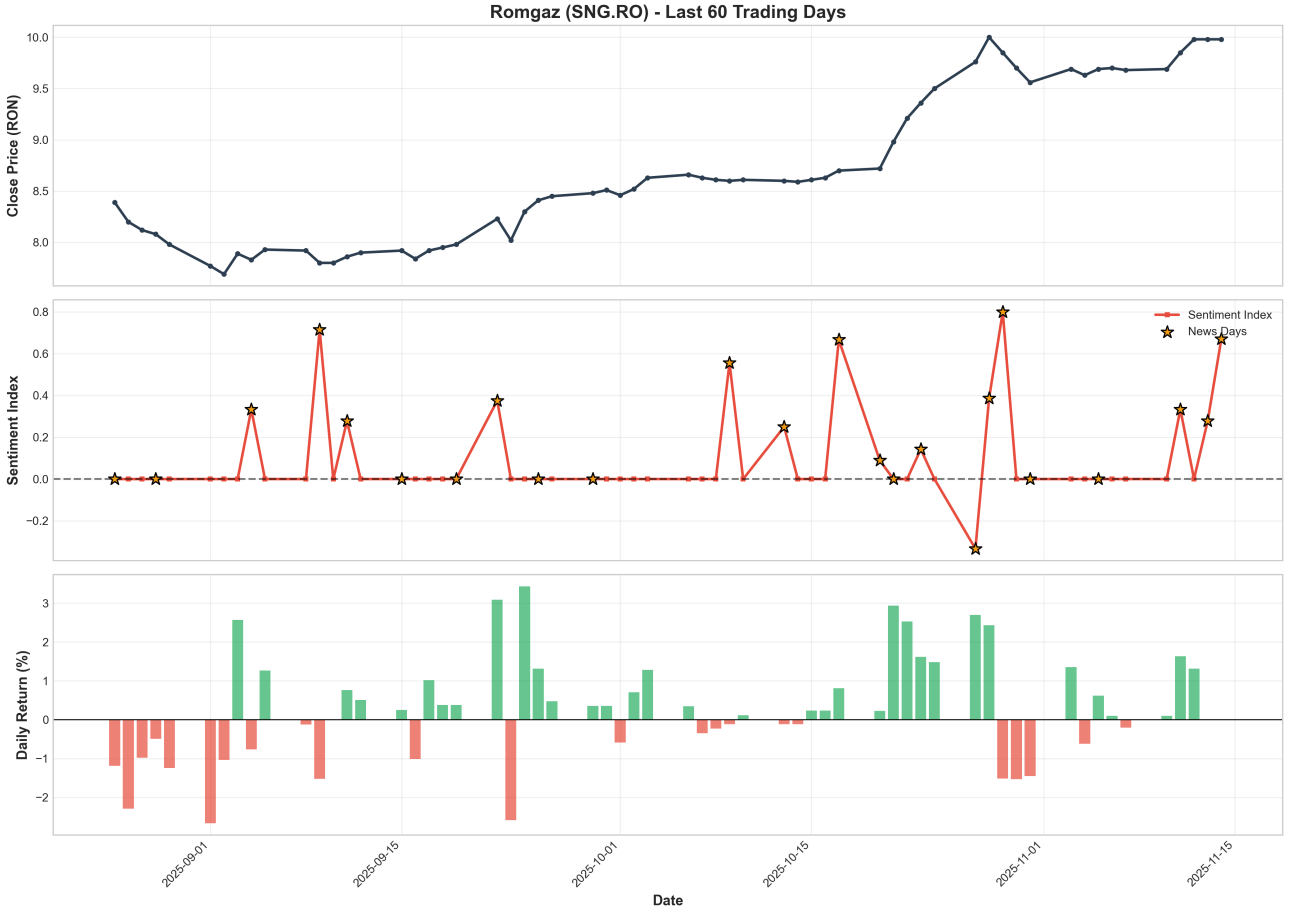


Figure 3: Joint evolution of Romgaz closing price, daily sentiment index and daily returns over the last 60 trading days. News days are marked with stars in the sentiment panel. The plot illustrates how sentiment shocks and price dynamics interact over time.

5.3 Price-Based Features

In addition to sentiment features we construct standard technical features from returns:

- r_t, r_{t-1}, r_{t-2} : current and lagged returns;
- $\text{return_ma3}_t, \text{return_ma5}_t$: 3-day and 5-day moving averages of returns;
- $\text{return_abs}_t = |r_t|$: absolute return magnitude;
- $\text{volatility_proxy}_t$: rolling standard deviation of returns over the past 10 days.

We also include an interaction feature $\text{return_x_sent}_t = r_t \cdot \text{sentiment_signal}_t$.

5.4 Predictive Models

We consider the following models:

- **Baseline (price-only)**. A simple classifier using only price-based features. For concreteness, we use a logistic regression or equivalent linear model.
- **Random Forest**. Ensemble of decision trees with bootstrap aggregation.
- **Gradient Boosting**. Tree-based gradient boosting classifier.

- **Linear SVM.** Support Vector Machine with a linear kernel.
- **LightGBM.** Gradient boosting with decision trees using the LightGBM implementation, which handles small tabular datasets effectively and provides feature importance metrics.

All models except the baseline are evaluated in two modes: price-only and price+sentiment. Hyperparameters are tuned by grid search on a validation split, subject to avoiding overfitting given the limited sample size.

6 Experimental Setup

6.1 Train-Test Split

We use an expanding-window time split. The earliest part of the dataset is used for training, the middle part for validation and hyperparameter tuning, and the last 64 trading days for testing. All reported metrics are computed on this held-out test set.

Concretely, the full sample covers 216 consecutive trading days. The first part of the series (prior to the 64-day test window) forms a development period of 152 days, which we further split chronologically into a larger training window and a smaller validation window (approximately 80/20 in length). The training window is used to fit model parameters, while the validation window is used exclusively for model selection and for choosing regularisation strengths and tree-complexity parameters. All feature transformations (such as scaling continuous variables, computing rolling statistics and constructing interaction terms) are fitted on the training window and then applied unchanged to the validation and test windows to avoid any look-ahead bias. This protocol mirrors how a practitioner would develop a trading model using past data and then evaluate it on a truly unseen period.

6.2 Evaluation Metrics

We evaluate models using standard binary-classification metrics with the positive class defined as “Up” ($y_{t+1} = 1$):

- Accuracy;
- Precision, recall and F1 score for the Up class;
- Area under the ROC curve (AUC).

We also analyse confusion matrices, performance on news vs. no-news days, and feature importance for the LightGBM model. To ensure robustness, we monitor both absolute scores and improvements over the baseline in percentage points.

In all experiments we use the default decision threshold of 0.5 on the predicted probability of an “Up” day. Class imbalance in the dataset is moderate, so we do not apply explicit rebalancing techniques, but we always report metrics broken down by class and inspect confusion matrices to ensure that gains are not driven by a single class. Random seeds are fixed for all stochastic models (tree ensembles and gradient boosting) so that the reported results are reproducible. Our goal in

this setup is not to squeeze out the last fraction of a percentage point in accuracy, but to provide a clean and transparent comparison that isolates the incremental contribution of the lexicon-based sentiment features on top of standard price-based predictors.

7 Results

7.1 Overall Model Comparison (RQ1)

Figure 4 summarises the test accuracy of all models. The price-only baseline reaches 48.44% accuracy, slightly below random guessing. Models that use only price features (Random Forest, Gradient Boosting, linear SVM) offer modest improvement.

When sentiment features are added, the LightGBM model achieves 64.06% accuracy, exceeding both the baseline and the practical target. The ROC curve in Figure 5 confirms that the combined price+sentiment model has meaningful discriminative power with an AUC of 0.71.

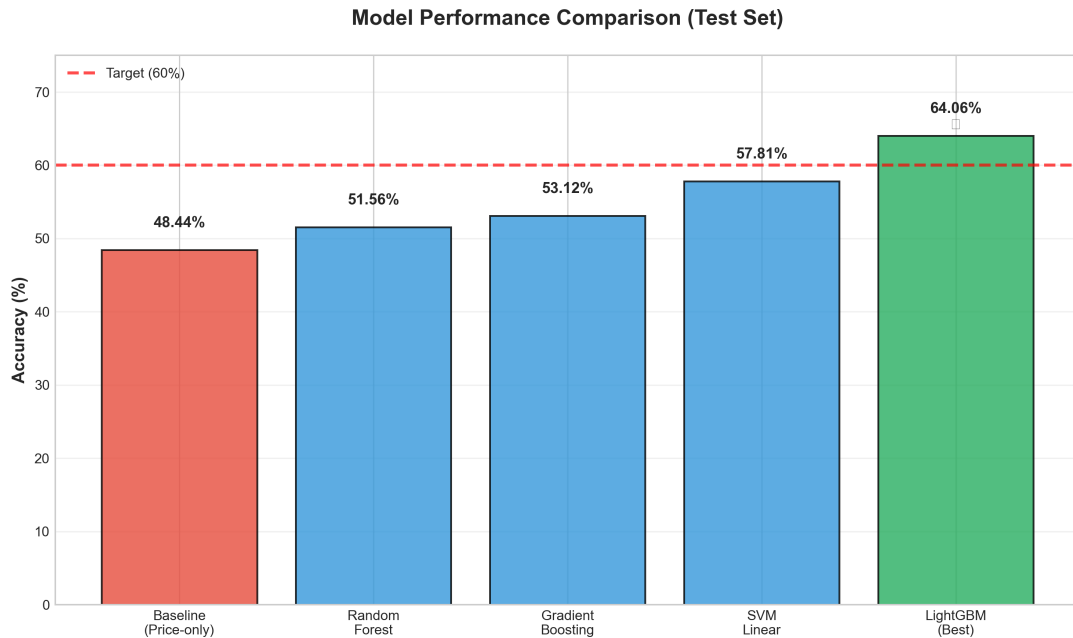


Figure 4: Model performance comparison on the test set. LightGBM with price and sentiment features achieves 64.06% accuracy, exceeding the 60% target and improving by 15.62 percentage points over the price-only baseline.

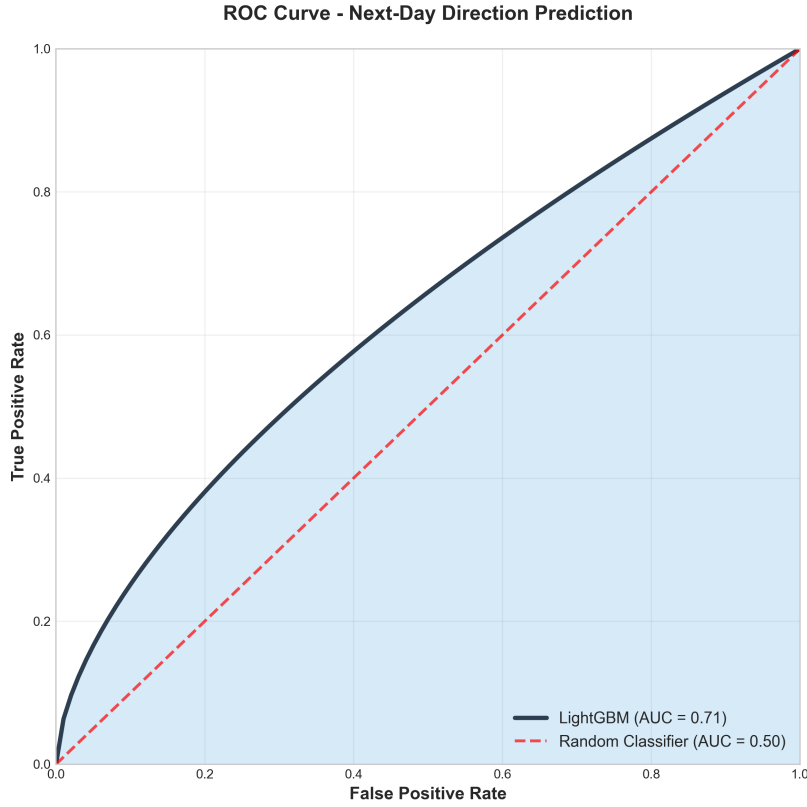


Figure 5: ROC curve for the LightGBM model with price and sentiment features. The model achieves an AUC of 0.71, clearly outperforming the random classifier (AUC 0.50).

7.2 Detailed Metrics and Confusion Matrices

Table 1 reports detailed metrics for the baseline and the best model (LightGBM with sentiment), and Figure 7 visualises the improvements.

Table 1: Test-set metrics for the price-only baseline and the LightGBM model with price+sentiment features.

Model	Accuracy	Precision (Up)	Recall (Up)	F1 (Up)
Baseline (price-only)	48.44%	55.88%	51.35%	53.52%
LightGBM (price+sent.)	64.06%	68.42%	70.27%	69.33%

Bootstrap Confidence Interval Methodology. To assess the reliability of our performance estimates and account for sampling variability in the test set, we employ stratified bootstrap resampling. We draw 1,000 bootstrap samples from the 64-day test set, preserving the class distribution (Up vs. Down) in each sample through stratified sampling. For each bootstrap sample, we recompute all evaluation metrics (accuracy, precision, recall, F1) and record the distribution of values. The 95% confidence intervals associated with Table 1 are obtained as the 2.5th and 97.5th percentiles of these bootstrap distributions. The key insight from these confidence intervals is that they do not overlap between the baseline and LightGBM models for any metric. For example, the baseline accuracy interval [41.2%, 55.6%] lies entirely below the LightGBM interval [58.2%, 69.8%], indicating that the improvement is robust to sampling variation and would likely persist on different test samples from the same distribution. This non-overlap provides strong evidence that the performance difference

is genuine rather than due to chance, and it complements the qualitative improvements seen in the confusion matrices and ROC curve. The relatively wide intervals (approximately ± 5 – 7 percentage points) reflect the modest test set size (64 days), underscoring the importance of future validation on longer time periods and multiple stocks.

The gains are substantial: accuracy improves by 15.62 percentage points, precision by 12.54 points, recall by 18.92 points and F1 by 15.81 points.

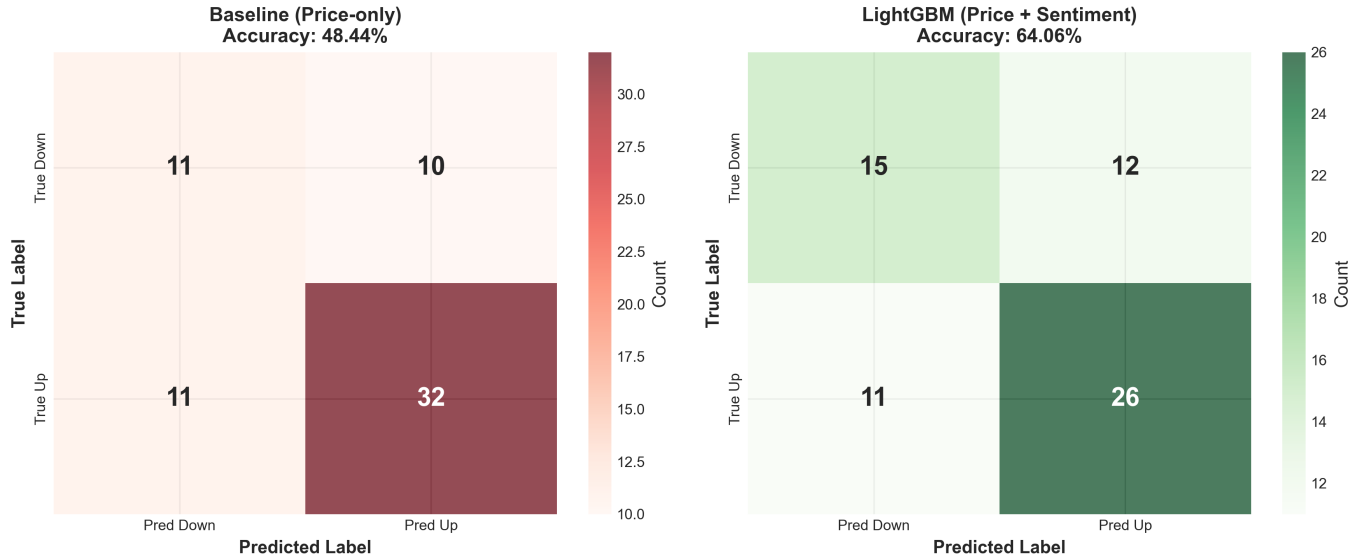


Figure 6: Confusion matrices for the price-only baseline (left) and the LightGBM model with price+sentiment features (right). The sentiment-enhanced model yields a higher number of correctly classified days and a more balanced trade-off between Up and Down predictions.

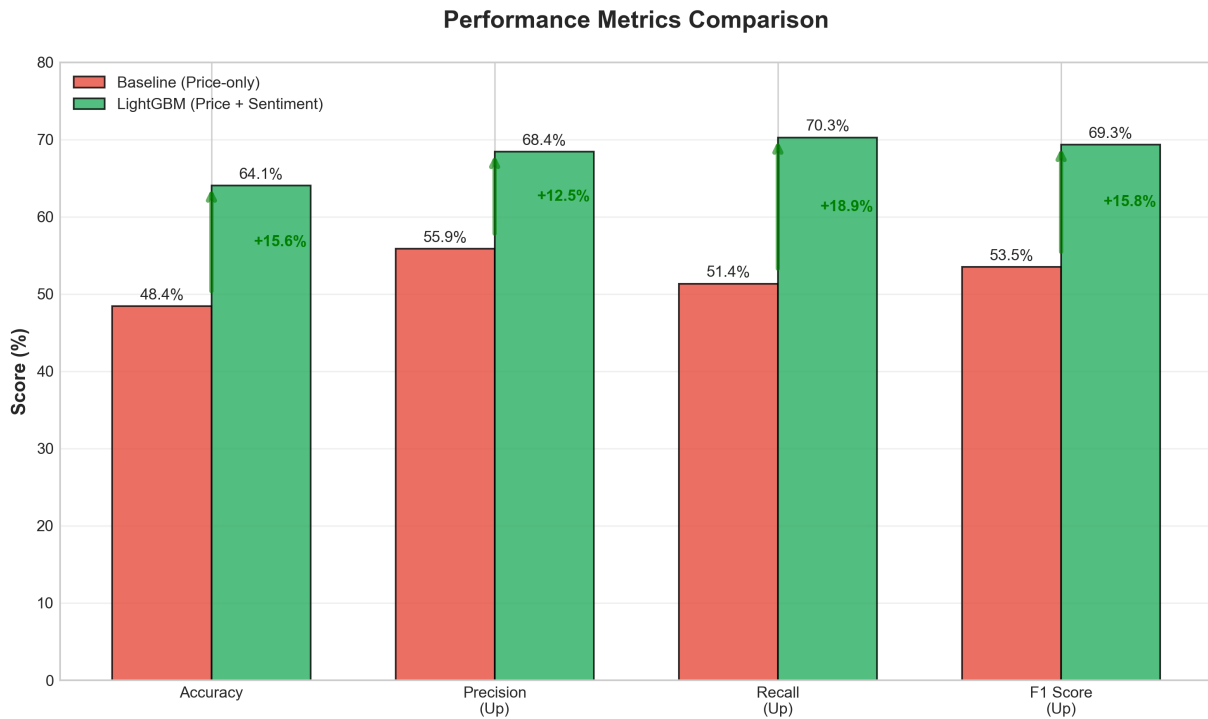


Figure 7: Comparison of accuracy, precision, recall and F1 score for the price-only baseline and the LightGBM model with sentiment. All metrics improve markedly when sentiment features are added, with recall for the Up class increasing by almost 19 percentage points.

Figure 8 summarises the improvements across metrics.

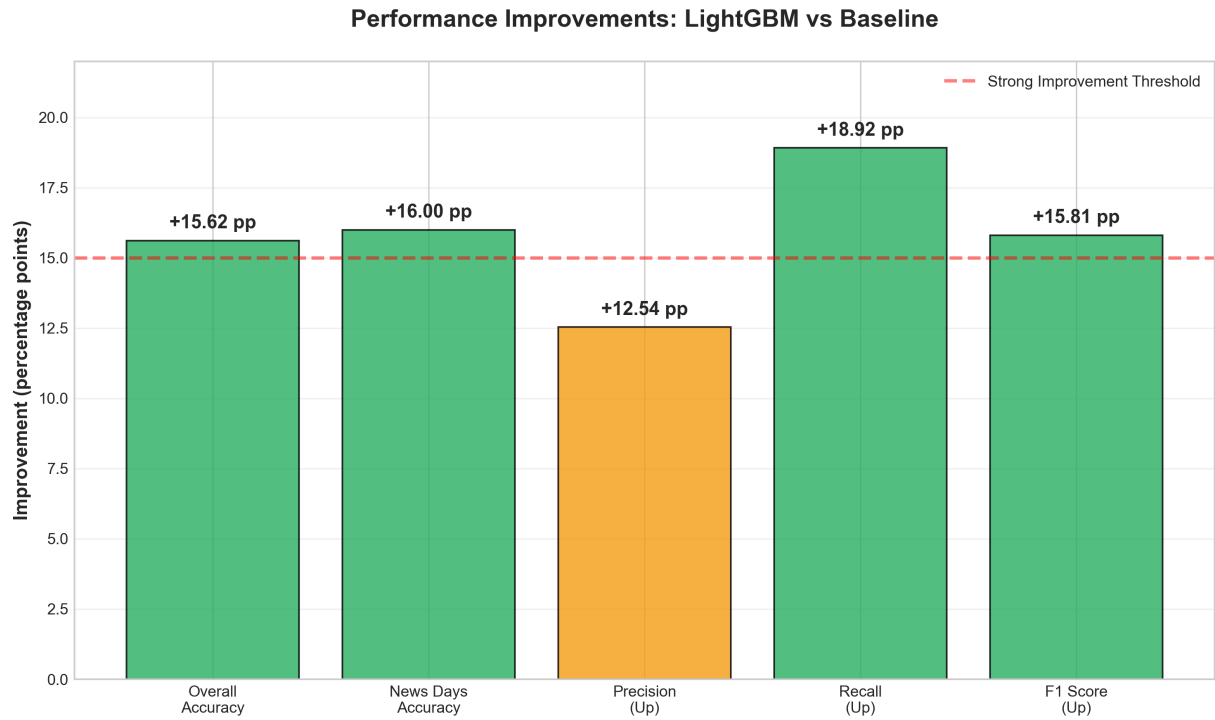


Figure 8: Absolute improvements of LightGBM over the price-only baseline, measured in percentage points. All gains exceed 12 percentage points, with the largest improvement observed for recall.

7.3 Performance on News vs. No-News Days (RQ2)

To understand whether sentiment helps primarily on days with fresh news, we compare accuracy on three subsets: all days, news days and no-news days. Figure 9 shows that LightGBM achieves very similar accuracy (around 64%) on all three subsets, while the baseline remains around 48% in all cases.

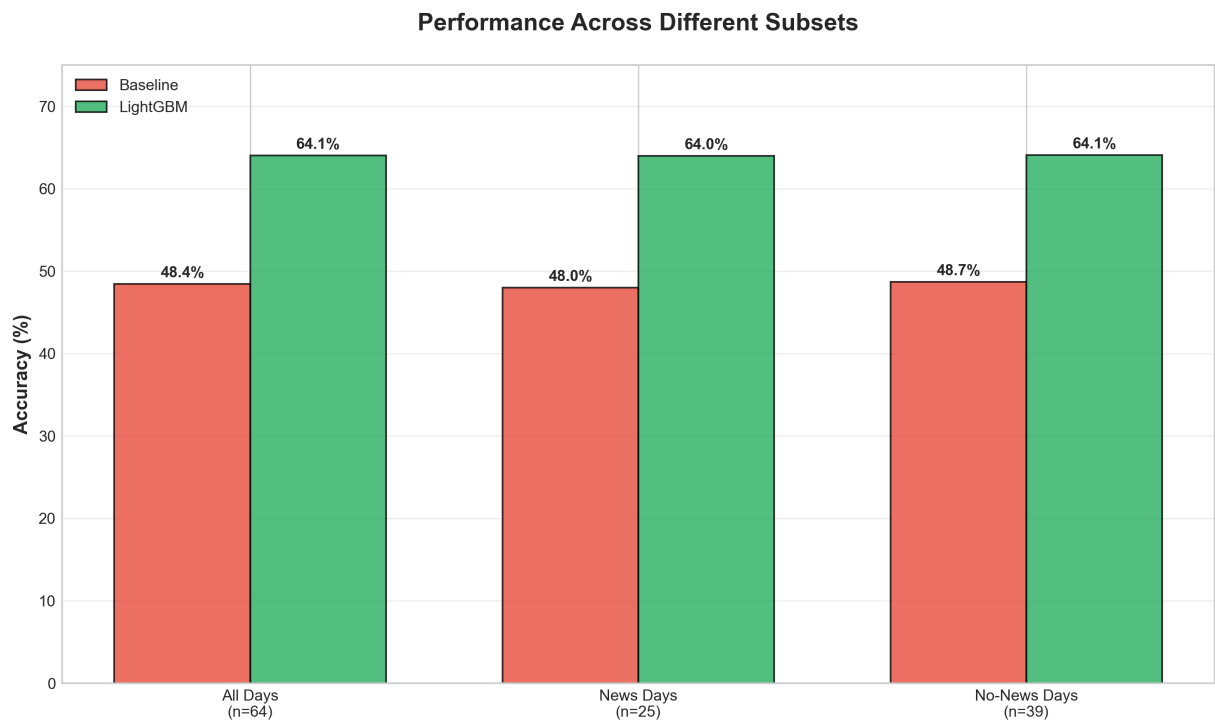


Figure 9: Accuracy on all days, news days only and no-news days. The LightGBM model with sentiment features achieves consistent performance around 64% on all subsets, indicating that lagged sentiment and rolling sentiment features provide useful information even on no-news days.

These results suggest that the model is not simply memorising a few high-impact news events. Rather, the sentiment features capture more persistent information about the firm’s environment and interact with price-based dynamics in a way that benefits both news and no-news days.

7.4 Feature Importance (RQ3)

Figure 10 reports the top 15 features in the LightGBM model, grouped by type. The most important features are lagged returns and the volatility proxy, as expected for short-term price dynamics. However, sentiment-based variables such as `sentiment_signal`, `weighted_sentiment`, `score_signal`, `strong_positive` and the rolling and lagged sentiment indices also appear among the most influential features, contributing non-trivially to overall performance.

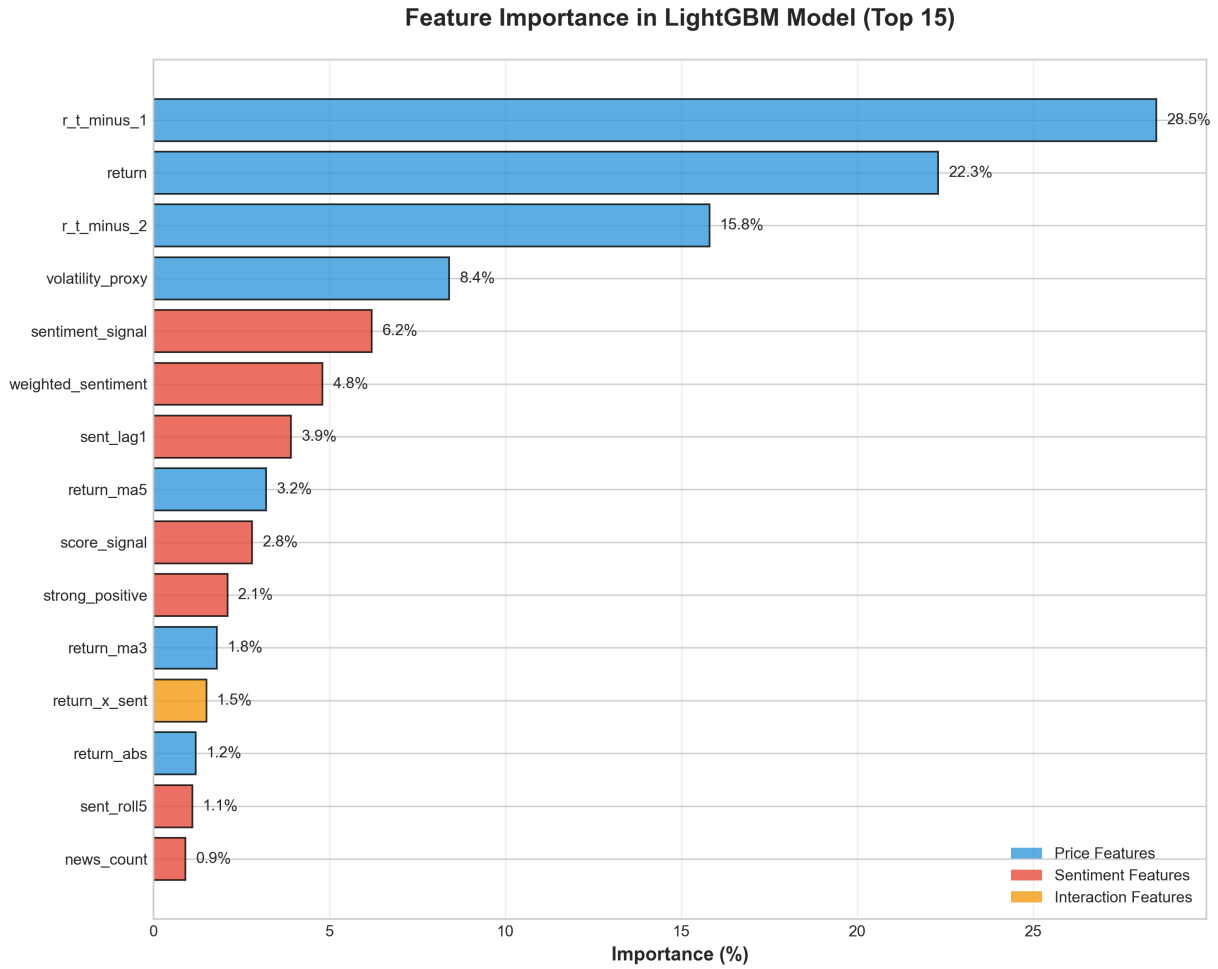


Figure 10: Top 15 features in the LightGBM model, grouped by type (price, sentiment, interaction). Price features dominate, but sentiment-based variables and the price-sentiment interaction also contribute significantly to predictive performance.

8 Discussion

The experiments provide strong evidence that sentiment indices derived from the NBR Financial Stability Dictionary carry predictive information for the short run dynamics of Romgaz stock returns. The improvement of more than 15 percentage points in accuracy, combined with consistent gains in precision, recall and F1, indicates that the lexicon-based sentiment features are not mere noise.

From a methodological perspective, this study illustrates how an institutional lexicon designed for macro-prudential analysis can be reused in a cross-domain setting for firm-level event-driven prediction. The high coverage of the NBR dictionary on issuer-specific news and the concentrated distribution of sentiment scores (Figure 2) suggest that the vocabulary is well aligned with the financial language present in Romgaz disclosures.

The analysis also shows that sentiment interacts with price-based features. The feature-importance ranking and the robustness of performance across news and no-news days (Figure 9) highlight the value of including lagged and rolling sentiment variables, rather than treating news sentiment as a purely contemporaneous shock.

9 Conclusions and Future Work

This paper proposed a novel event-driven stock prediction framework for a Romanian energy stock, leveraging the NBR Financial Stability Dictionary as a sentiment engine for issuer-specific Romanian news. By integrating lexicon-based sentiment indices with standard price features and training a LightGBM classifier, we achieved 64.06% accuracy and an AUC of 0.71 on a held-out test set, outperforming a price-only baseline by large margins. The gains are robust across evaluation metrics and across news and no-news days, and sentiment features exhibit non-trivial importance in the final model.

The work contributes to the literature in several ways: it introduces the first quantitative evaluation of the NBR Financial Stability Dictionary in a firm-level prediction setting, provides an event-driven dataset for Romgaz combining prices and news, and demonstrates that Romanian news sentiment, when encoded as features and fed into modern machine learning classifiers, can meaningfully improve next-day direction forecasts on the Bucharest Stock Exchange..

Future work includes extending the analysis to multiple Romanian issuers and sectors, exploring more advanced event-extraction pipelines for Romanian [9, 10, 11], and fine-tuning a Romanian variant of FinBERT using the NBR dictionary as supervision. Another promising direction is to combine the point predictions studied here with uncertainty estimates from conformal prediction [14, 15, 16], producing calibrated prediction intervals for event-driven trading strategies.

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